Assignment # 03

Title: Practice Problems

Course title: Introduction to Probability and Statistics

Course code: STAT 251

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Submitted to-

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Q1) Given, $n = 36$

$$\Sigma x_i = 6.72 + 6.77 + 6.82 + 6.70 + 6.78 + 6.70 + 6.62 + 6.75 + 6.66 + 6.66 + 6.64 + 6.76 + 6.73 + 6.80 + 6.72 + 6.76 + 6.76 + 6.68 + 6.66 + 6.62 + 6.72 + 6.76 + 6.70 + 6.78 + 6.78 + 6.67 + 6.70 + 6.72 + 6.74 + 6.81 + 6.79 + 6.78 + 6.66 + 6.76 + 6.76 + 6.82$$

$$= 242.814$$

a) sample mean, $\bar{x} = \frac{\Sigma x_i}{n}$

$$\Rightarrow \bar{x} = \frac{242.14}{36}$$

$$= 6.726$$

b) Sample standard Deviation,

$$s = \sqrt{\frac{\Sigma_{i=1}^{n} (x_i - \bar{x})^2}{n-1}}$$

x_i	f_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$f_i(x_i - \bar{x})^2$
6.62	2	−0.1061	0.011257	0.022514
6.64	1	−0.0861	0.007413	0.007413
6.66	4	−0.0661	0.004369	0.017476
6.67	1	−0.0561	0.003197	0.003197
6.68	1	−0.0461	0.002125	0.002125
6.70	4	−0.0261	0.000681	0.002724
6.72	5	−0.0061	0.000037	0.000185
6.73	1	0.0039	0.000015	0.000015
6.74	1	0.0139	0.000193	0.000193
6.75	1	0.0239	0.000571	0.000571
6.76	2	0.0339	0.001149	0.002298
6.77	1	0.0439	0.002505	0.002505
6.79	1	0.0639	0.004083	0.004083
6.80	1	0.0739	0.005461	0.005461
6.81	1	0.0839	0.007039	0.007039
6.82	1	0.0939	0.008817	0.008817
$n =$	36			$= 0.100447$

$$\therefore S = \sqrt{\frac{0.100447}{36-1}}$$

$$= \boxed{0.054}$$ Ans:

b) $n = 36$

max $= 6.82$

min $= 6.62$

Range $= 6.82 - 6.62 = 0.2$

number of bucket, $k = \lceil \log_2 n + 1 \rceil$

$= \lceil \log_2 36 + 1 \rceil$

$= \lceil 3.322 \log_{10} 36 + 1 \rceil$

$= \lceil 6.17 \rceil$

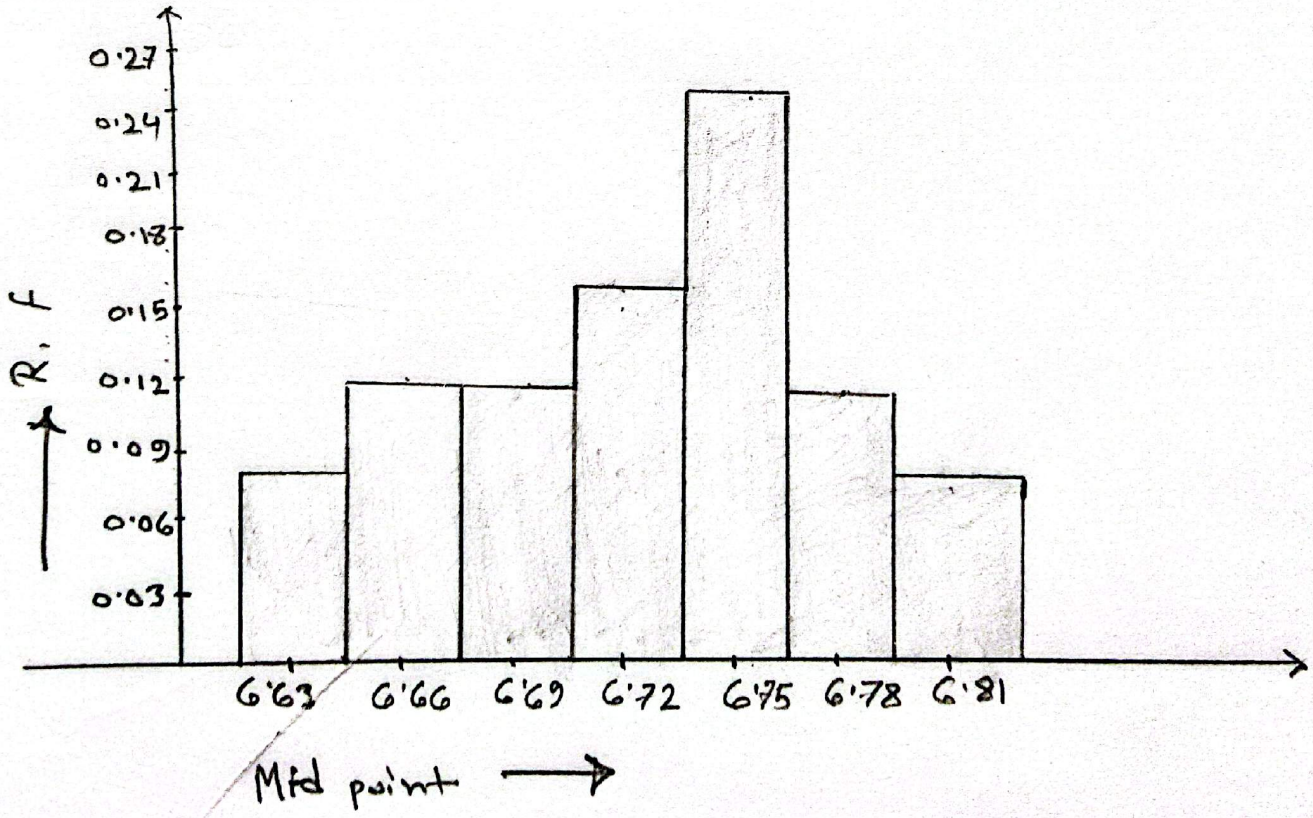
$= 7$

interval, $h = \frac{R}{k} = \frac{0.2}{7}$

$= 0.0285$

≈ 0.02

class Limits	Midpoint	frequency (fi)	Relative Frequency fi/n
6·62–6·64	6·63	III (3)	0·0833
6·65–6·67	6·66	IIII (5)	0·1389
6·68–6·7	6·69	IIII (5)	0·1389
6·71–6·73	6·72	IIII I (6)	0·1667
6·74–6·76	6·75	IIII IIII (9)	0·2500
6·77–6·79	6·78	IIII (5)	0·1389
6·80–6·82	6·81	III (3)	0·0833
~~6·81–6·83~~	~~6·82~~	n = 36	



Semi-Bell shaped histogram.

(c) It is observed from the drawn histogram that it is roughly symmetric, tapering off on both sides. It has slight tail fail.

So, therefore it is approximately bell-shaped.

Q2 :

a) We have to find P(Tested ∩ Failed)

Step-1 :

$P(\text{Pass All}) = (1-0.01)(1-0.03)(1-0.02)(1-0.01)$

$= 0.99 * 0.97 * 0.98 * 0.99$

$= 0.9318$

$P(\text{Failed} \mid \text{Tested}) = 1 - P(\text{All Pass All})$

$= 1 - 0.9318$

$= 0.0682$

Step-2 :

$P(\text{Tested and Failed}) = \frac{1}{4} \times 0.0682$

$= 0.01705$

Therefore, Ans : $(P \text{ Tested} \cap \text{Failed})$

$= 0.01705$

$= 1.705\%$

b) We have to find $P((F_2 \cup F_3) \mid \text{Tested})$

$$P(F_2 \cup F_3) = P(F_2) + P(F_3) - P(F_2 \cap F_3)$$
$$= 0.03 + 0.02 - 0.03 * 0.02$$
$$= 0.0494$$

Therefore, Ans: 0.0494 or **4.94 %**

c) A CD is rejected if it is tested and Failed.

From part-a, we got $P(\text{Rejected})$

$$P(\text{Rejected}) = 0.01705$$

To find the expected number in sample of 100 : $E = n * P(\text{Rejected})$

$$= 100 * 0.01705$$
$$= 1.705$$

Therefore, Ans: **1.705 CDs**

(d)

From the Bayes' Theorem,

~~P(Tested/Defective) =~~

$$P(\text{Tested}/\text{Defective}) = \frac{P(\text{Defective}/\text{Tested}) * P(\text{Tested})}{P(\text{Defective})}$$

$P(\text{Defective} / \text{Tested}) = 0.0682$ (part a)

$P(\text{Tested}) = 0.25$

$P(\text{Defective}) = P(\text{Tested} \cap \text{Failed}) = \frac{\text{~~0.01705~~}}{0.0682}$

Therefore,

$$P(\text{Tested}/\text{Defective}) = \frac{0.0682 * 0.25}{\frac{\text{~~0.01705~~}}{0.0682}}$$

$$= 0.25$$

Q3

(a) We have to find,

- P(union | new company, same field) = ?
- From the stem, we get,
  New Company (same field) : 13 union + 10 non union
  total = 23
- Therefore, P(union | new company) = $\frac{13}{23}$ = 0.565

(b) We have to find,

- P(unemployed 1 year | union member)
- Given, union colum total = 59, of which 2are unemployed.
- Therefore, P(unemployed | union) = $\frac{2}{59}$ = 0.0339